

Statement of Research Interests

My interest in machine learning began with a simple question: how can intelligent systems learn to make effective decisions in complex environments? This question has guided my academic and research journey and led me to focus on deep reinforcement learning and explainable artificial intelligence.

I am currently completing my MSc in Computer and Systems Engineering at Minia University, where I have been working as a Research Assistant in the Department of Computer and Systems Engineering since 2020. My research has primarily focused on deep reinforcement learning and optimal control for autonomous driving applications. Through my master's thesis on motion planning for autonomous vehicles, I have investigated how learning-based methods can be combined with control-theoretic principles to solve challenging decision-making problems.

A significant part of my research has involved studying the impact of hyperparameters on reinforcement learning performance and learning dynamics. This work exposed me to the practical challenges of training deep learning systems and sparked my interest in understanding not only whether a model works, but also why it works. As a result, I became increasingly interested in explainable AI and the broader question of how learned representations influence model behavior. More generally, I am interested in understanding how intelligent agents learn, adapt, reason and make decisions over time, and how such systems can be made more interpretable, controllable, and reliable.

Beyond research, my teaching activities in machine learning, programming, control theory, systems analysis, and cyber-physical systems have strengthened my analytical and communication skills. These experiences have reinforced my desire to pursue a research career and contribute to the advancement of intelligent learning systems.

My long-term goal is to conduct research that improves our understanding of how learning systems operate and how they can be made more capable, interpretable, and trustworthy. I am excited by the possibility of pursuing these goals within your group and contributing to its research community.